BreathTwin: Multimodal Asthma Flare-Up Digital Twin

Forecasting Acute Asthma Exacerbations 24 to 48 Hours Ahead in High-Pollution Indian Cities

// PROJECT SUMMARY

BreathTwin fuses longitudinal Electronic Health Records (EHR), 20-channel continuous wearable telemetry, and real-time Central Pollution Control Board (CPCB) air quality data into a patient-specific Digital Twin. A mechanistic latent airway inflammation ODE coupled with a 3-stage neural-tree ensemble predicts 24-hour and 48-hour exacerbation risk, explains the physiological drivers, and simulates counterfactual interventions before a clinician advises the patient.

// TEAM & SUBMISSION COMPLIANCE

Team Name: Team DigiTwin (BreathTwin) | Lead: Aarush Bhattacharjee | Contributors: Shubhang, Rakshit Gupta

Domain: Pulmonary Informatics & Clinical Decision Support | License: MIT Open-Source License

Data Policy: 100% Synthetic Patient Cohort + Open Public CPCB / Open-Meteo Environmental Streams

Closing the 24-36 Hour Sub-Clinical Inflammatory Blindspot

Shifting Asthma Care from Reactive Emergency Visits to Proactive Pre-Symptomatic Triage

// THE CLINICAL PROBLEM

1. High Mortality Burden: India accounts for >42% of global asthma deaths (~13% of global prevalence).
2. Invisible Deterioration: Routine pulmonology visits occur months apart. Acute PM2.5 smog surges trigger bronchial mucosal swelling and smooth-muscle constriction 24 to 36 hours before the patient perceives dyspnea.
3. Fragmented Signals: Static spirometry, wearable sensors, and tomorrow's AQI exist in isolated silos.

// HEALTHCARE USE CASE (DOCTOR & PATIENT WORKFLOWS)

1. Cohort Triage: Pulmonologist opens the ranked patient list sorted by 48-hour exacerbation risk.
2. Explainable Inspection: Doctor inspects 90% conformal risk intervals, SHAP attributions, and 3D lung state.
3. Counterfactual What-If Testing: Doctor tests 'Stay Indoors', 'Wear N95 Mask', 'HEPA Filter', or 'ICS Controller Adherence Step-Up' on the patient's digital twin to quantify risk reduction prior to care.
4. Patient Companion Portal: Patient logs daily symptoms/inhaler use and receives plain-language precautions.

Hybrid Three-Stage Multimodal Ensemble + Latent Airway ODE

Combining Physiological Differential Equations with Deep Temporal Convolutions and Gradient Boosting

// STAGE 1 & STAGE 2 PREDICTIVE ENSEMBLE

Stage 1A - Dilated TCN (DigitalTwinFusionModel): StaticMLPEncoder (60 EHR features -> 64D) + Conv1DDynamicEncoder (3 dilated 1D conv blocks, d=1,2,4 over 20x12h telemetry -> 128D).

Stage 1B - XGBoost Multi-Target Suite: Rolling window statistics (mean, std, min, max, delta) + lag shifts.

Stage 2 - Stacking Meta-Learner (RespiTwinMLPMetaLearner): PyTorch MLP (12D -> 32D -> 16D) combining Stage-1 probabilities into calibrated 24h risk, 48h risk, and 4-class exacerbation severity (Grade 0-3).

// MECHANISTIC ODE SIMULATOR & STAGE 3 CLINICAL COPILOT

Latent Airway Inflammation ODE: dI/dt = -k*I + a*Dose_PM(t) + b*Trigger(t) + d_sys - c*Controller(t) + sigma*xi(t) Integrated via Euler-Maruyama (dt=1.0h) to track mucosal inflammation I(t) and power What-If simulations.

Stage 3 Clinical Copilot & NLP Extractor: Pydantic-validated LLM reports (GPT-4o / Llama-3.1 / Qwen2.5) and free-text clinical note perturbation engine (Typhoid, Dengue, Dehydration -> vital & ODE modifiers).

Validation Results, Ablation Studies, and Operational Safety

Evaluated on Held-Out Synthetic Cohort Split by Patient (800 Train / 100 Val / 100 Test)

// HELD-OUT TEST METRICS & BENCHMARK COMPARISON

Stacked Meta-Learner (TCN + XGBoost + MLP): 24h ROC-AUC = 0.9399 | 48h ROC-AUC = 0.9428 | Severity F1 = 0.8435

24h Precision @ 0.50 = 0.8560 | 24h Recall @ 0.50 = 0.7281 | Mean Warning Lead Time = 42.4 +/- 4.1 hours

BreathTwin Hybrid vs Baselines: Outperforms XGBoost-only (PR-AUC 0.614, Lead 18.6h) and LSTM (PR-AUC 0.671, Lead 22.1h)

End-to-End Counterfactual Inference Latency: < 65 ms per 48-hour trajectory simulation.

// ETHICAL GUARDRAILS & OPERATIONAL BOUNDARIES

1. Synthetic Data Only: Zero real patient PII/PHI; designed in alignment with DPDP Act data minimization.
2. Decision Support, Not Diagnosis: Estimates exacerbation risk to assist triage; never issues prescriptions.
3. Reproducible Open-Source Stack: Docker Compose PostgreSQL 16 + FastAPI + Next.js 16 under MIT License.